

BU283

Chapter 4 Notes

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CH.4
Annuities and Loans

Chapter 3 moved ONE cash flow through time. Chapter 4 moves a whole STREAM of equal payments at once. The trick is the same: bring everything to one date, then compare.

4.1 Future Value of an Annuity

DEF Annuity

A series of EQUAL payments made at EQUAL intervals. Can be inflows or outflows.

DEF Ordinary annuity

Payments at the END of each period. This is the default unless told otherwise.

DEF Annuity due

Payments at the BEGINNING of each period (rent, leases, tuition).

FORMULA FV of an ordinary annuity

$$FV_n = PMT\left[\frac{(1+i)^n - 1}{i}\right]$$

The bracket is the FVIFA. n = number of payments.

EXAMPLE $100 per year for 3 years at 10%

Brute force: $100(1.10)^2 + 100(1.10) + 100 = 121 + 110 + 100 = \$331$
Formula: $100 \times \frac{1.10^3 - 1}{0.10} = 100 \times 3.31 = \$331$

EXAMPLE $50,000 per year at 8% for 20 years

$$FV = 50000 \times \frac{1.08^{20} - 1}{0.08} = \$2,288,098.21$$

Calculator: N 20, I/Y 8, PV 0, PMT -50000, CPT FV.

FORMULA Solving for the payment (savings goal)

$$PMT = \frac{FV_n}{\left[\frac{(1+i)^n - 1}{i}\right]}$$

EXAMPLE Save $1,000,000 in 30 years at 8%

$$PMT = \frac{1000000}{113.28} = \$8,827.43 \text{ per year}$$

FORMULA FV of an annuity due

$$FV_n^{due} = PMT\left[\frac{(1+i)^n - 1}{i}\right](1+i)$$

Each payment gets one extra period of interest, so multiply by $(1+i)$.

EXAMPLE $5,000 per year for 6 years at 7%, annuity due

Ordinary: $5000 \times 7.1533 = \$35,766.45$
Due: $35,766.45 \times 1.07 = \$38,270.11$

WATCH OUT

Calculator for annuity due: 2ND, BGN, 2ND, SET. The screen shows BGN. Switch it back to END when you're done or every answer after is wrong.

4.2 Present Value of an Annuity

FORMULA PV of an ordinary annuity

$$PV_0 = PMT\left[\frac{1 - \frac{1}{(1+i)^n}}{i}\right]$$

The bracket is the PVIFA.

EXAMPLE $100 per year for 4 years at 10%

$$PV = 100 \times \frac{1 - 1.10^{-4}}{0.10} = 100 \times 3.1699 = \$316.99$$

Calculator: N 4, I/Y 10, PMT 100, FV 0, CPT PV.

EXAMPLE Solve for the rate

You have $562,889 and want to withdraw $50,000 a year for 30 years. What return do you need?
N 30, PV 562889, PMT -50000, FV 0, CPT I/Y $= 8\%$.

WATCH OUT

PV and PMT need opposite signs. You put money IN (PV) and take money OUT (PMT).

FORMULA PV of an annuity due (Eq. 4.6)

$$PV_0^{due} = PMT\left[\frac{1-\frac{1}{(1+i)^n}}{i}\right](1+i)$$

EXAMPLE $100 at times 0, 1, 2, 3 at 10%

Ordinary PV of 4 payments is \$316.99, so due $= 316.99 \times 1.10 = \$348.69$

DEF Perpetuity

An annuity that pays forever.

FORMULA PV of a perpetuity

$$PV_0 = \frac{PMT}{i}$$

First payment is one period from now.

EXAMPLE $100 forever at 10%

$PV = 100/0.10 = \$1,000$

4.3 Advanced TVM Problems

DEF Equation of value

Cash inflows = cash outflows, once everything is moved to the SAME point in time.

DEF Focal date

The common date you move every cash flow to. Pick the one that makes the math easiest.

DEF Game plan

1. Write down what you know.
2. Draw a timeline.
3. Pick a focal date.
4. Label each cash flow as a lump sum or an annuity.
5. Check compounding frequency and whether payments are beginning or end of period.

DEF Imbedded annuity

An annuity sitting inside a bigger stream of cash flows.

DEF Deferred annuity

Equal payments that start at some future date instead of right away.

WATCH OUT

The annuity formula gives you the value ONE PERIOD BEFORE the first payment. Payments starting at time 2 give a PV at time 1, not time 0.

EXAMPLE Deferred annuity: $100 at times 2, 3, 4 at 10%

Step 1, annuity value at time 1: $100 \times 2.4869 = \$248.69$
Step 2, discount to time 0: $248.69/1.10 = \$226.08$

EXAMPLE Two annuities at 10%

$200 at times 1 to 3, plus $100 at times 4 to 6.
First: $200 \times 2.4869 = \$497.37$
Second: $100 \times 2.4869 = 248.69$ at time 3, then $248.69/1.10^3 = \$186.84$
Total PV $= \$684.21$

EXAMPLE Retirement: save for 30 years, spend for 30

Save $X per year at 5%, then spend $100,000 per year at 3%. End of period. Focal date = retirement (year 30).

$$X \cdot FVIFA_{5\%,30} = 100000 \cdot PVIFA_{3\%,30}$$

$$X \times 66.439 = 1,960,044 \quad \Rightarrow \quad X = \$29,501.48$$

EXAMPLE Retirement, monthly, with an inheritance

Save monthly for 30 years at 5%. Spend $10,000 a month for 35 years at 3%. Leave $500,000 at death. Focal date = retirement.

$$X \cdot FVIFA_{\frac{0.05}{12},360} = 10000 \cdot PVIFA_{\frac{0.03}{12},420} + \frac{500000}{\left(1+\frac{0.03}{12}\right)^{420}}$$

$$\text{Need at retirement} = \$2,773,612 \quad \Rightarrow \quad X = \$3,332.63 \text{ per month}$$

4.4 Balloon Loans

DEF Balloon loan

Zero or low payments while the loan is outstanding, with the remaining balance due in one lump at maturity.

DEF Amortized loan

Level, regular payments that fully pay off the loan. Each payment covers interest first, then the rest reduces principal.

FORMULA Balloon payment

$$\text{Balloon} = P(1+i)^n - PMT\left[\frac{(1+i)^n - 1}{i}\right]$$

What the loan grew to, minus what your payments grew to. Same as the FV key on the calculator.

EXAMPLE Borrow $20,000 at 8%, pay $3,000 a year for 5 years

$$\text{Balloon} = 20000(1.08)^5 - 3000(5.8666) = 29,386.56 - 17,599.80 = \$11,786.76$$

Calculator: N 5, I/Y 8, PV 20000, PMT -3000, CPT FV.

4.5 Amortized Loans

FORMULA Equation of value for a loan

$$\text{Principal} = PMT \times PVIFA_{i,n}$$

Payments are an ordinary annuity. Car loans and mortgages work this way.

EXAMPLE $190,000 house, 6%, 20 years, monthly

$$PMT = \frac{190000}{PVIFA_{0.5\%,\,240}} = \$1,361.22$$

This is how the slide solves it, with $j = 0.06/12$. See the mortgage section for the Bank Act version.

DEF Amortization schedule

A table showing each payment split into interest and principal, and the balance left after each one. Interest shrinks over time and principal grows.

FORMULA Each row of the schedule

$$\text{Interest}_t = \text{Balance}_{t-1} \times i \qquad \text{Principal}_t = PMT - \text{Interest}_t$$

New balance = old balance minus principal.

EXAMPLE $10,000, 5 years, 10%, annual payments

$PMT = 10000/PVIFA_{10\%,5} = \$2,637.97$

Yr	Interest	Principal	Balance
1	1,000.00	1,637.97	8,362.03
2	836.20	1,801.77	6,560.26
3	656.03	1,981.94	4,578.32
4	457.83	2,180.14	2,398.18
5	239.82	2,398.15	0.00

EXAMPLE Car loan: $82,900, 3.9% APR, 72 months, 13% tax

Tax goes into the loan: $82,900 \times 1.13 = \$93,677$
$PVIFA_{0.039/12,72} = 64.104$
$PMT = 93,677/64.104 = \$1,461.33$ per month

4.6 Car Leases

DEF Lease

Payments at the BEGINNING of each month (annuity due), plus a lump sum buyout at the end. You can pay the buyout and keep the car, or walk away.

FORMULA Equation of value for a lease

$$\text{Price} = PMT \times PVIFA^{due}_{j,nm} + \text{Buyout} \times PVIF_{j,nm}$$

Tax is added to each payment afterward: after-tax PMT = PMT $\times(1+T)$.

WATCH OUT

Always include the buyout, even if you plan to walk away. The left side is owning the car, so the right side must be too.

EXAMPLE $83,915 car, 8% APR, 48 months, $34,818 buyout, 13% tax

$j = 0.08/12$. $PVIFA^{due} = 41.235$. $PVIF = 0.7269$

$$PMT = \frac{83915 - 34818(0.7269)}{41.235} = \$1,421.25 \text{ pre-tax}$$

After tax: $1,421.25 \times 1.13 = \$1,606.01$

4.7 Mortgages and the Bank Act

DEF Mortgage

An amortized loan on a home. Principal outstanding = PMT $\times$ PVIFA.

DEF Bank Act rule

Canadian mortgage rates are compounded SEMI-ANNUALLY, no matter how often you pay. So you can NOT use $j = i/m$.

FORMULA Max EIR on a mortgage

$$EIR = \left(1 + \frac{i}{2}\right)^2 - 1$$

EXAMPLE Quoted mortgage rate 7.5%

$EIR = (1.0375)^2 - 1 = 7.64\%$

FORMULA Periodic mortgage rate j

$$j = \left(1 + \frac{i}{2}\right)^{2/m} - 1$$

m = payments per year: 12 monthly, 24 semi-monthly, 26 bi-weekly, 52 weekly.

EXAMPLE 6.05% quoted, semi-monthly payments

$$j = \left(1 + \frac{0.0605}{2}\right)^{2/24} - 1 = 0.002487$$

EXAMPLE Borrow $179,925 at 6.05%, 25 years, semi-monthly

$n \times m = 25 \times 24 = 600$ payments, $j = 0.002487$
$PVIFA = 311.53$
$PMT = 179{,}925/311.53 = \$577.54$

WATCH OUT

Mortgage = semi-annual compounding, use j. Car loans, leases, savings = use i/m. Read which one the question is.

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DEF Annuity keys

N = number of payments, I/Y = rate PER PERIOD in %, PMT = payment, PV and FV as usual.

DEF Annuity due

2ND, BGN, 2ND, SET. Back to END the same way.

DEF Amortization worksheet

2ND, AMORT. Set P1 and P2 to the payment range, then scroll down for BAL, PRN, INT.

WATCH OUT

For Bank Act mortgages, compute j first, then enter I/Y $= j \times 100$.

check Quick Questions

Q1

Why is an annuity due worth more than an ordinary annuity?

A. Every payment comes one period earlier, so each one earns (or is discounted) one less period. That's the extra $(1+i)$.

Q2

A deferred annuity's first payment is at time 5. The PVA formula gives you its value at what time?

A. Time 4. Then discount 4 more periods to get to time 0.

Q3

In an amortization schedule, why does the interest portion drop each payment?

A. Interest is charged on the remaining balance, and the balance falls every payment.

Q4

Why can't you use $j = 0.06/12$ for a Canadian mortgage at 6%?

A. The Bank Act makes mortgages compound semi-annually. Use $j = (1.03)^{1/6} - 1$ for

monthly payments.

Q5

Why must the lease equation include the buyout?

A. The price is the value of owning the car. Payments alone don’t get you ownership, so the buyout has to be on the right side too.